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PRIVATE SHIELD: Week 4 Update

I. INTRODUCTION

This weejs update covers the work completed after Previous weeks : data preparation and auditing, integration of a supplied model, and a trained MLP FedAvg checkpoint. The latest Dataset is the active source, with 46 features and 34 traffic categories. The earlier dataset workflow is retained only for reference.

II. IMPLEMENTATION WORK

A. Data Preparation and Audit

latest_audit.py now scans the active train, validation and test splits and reports exact float32 input overlaps. latest_pilot.py fits training-only scaling and exports client inputs, manifests, IDs and lineage. latest_pilot_runner.py coordinates audit and preparation with source checks and completion records.

B. Supplied Model Integration

The model and data versions of stacking.py recreate the supplied architecture, load its weights and reconstruct the missing preprocessing from training data. stacking_runner.py connects recovery and diagnostic inference. main.py adds explicit routing, while YAML files hold settings. This supplied stacking model remains separate from the trained FedAvg MLP.

D. N-BaIoT Model Experiments

n_balot_models.ipynb prepared 19,800 balanced records across 11 N-BaIoT traffic classes (115 features). Seven classifiers were compared for binary and multiclass detection using stratified 70/15/15 splits. The notebook also evaluated full-data binary Logistic Regression and class-weighted multiclass Logistic Regression, then packaged trained models with feature names, labels, requirements and prediction scripts.

C. FedAvg Training and Logging

The prepared pilot reuses the existing clients, server and sample-weighted FedAvg. An MLP with hidden layers of 64 and 32 units was trained for five rounds across three simulated clients, using 30,070 training records. The final-round model, update history, aggregation weights and validation logs were saved. A separate configuration evaluates the checkpoint without retraining.

III. VERIFICATION AND CURRENT STATUS

The full suite passed 777 cases. Reload checks confirmed changed weights, correct sample fractions and identical saved-model metrics. On 11,546 class-capped, overlap-filtered validation records, accuracy was 59.36% and macro F1 was 0.5019. Three classes still had zero recall. These are preliminary pilot results, not a final benchmark.

In separate N-BaIoT tests, full-data binary Logistic Regression reached 99.9764% accuracy on 203,660 test records. Balanced multiclass Logistic Regression reached 85.2862% on 2,970 records; LightGBM led multiclass validation (macro F1 0.998). These are non-federated notebook results.

FUTURE OBJECTIVES

- Agree the final split/grouping and non-IID client protocol, then extend minority-class evaluation and matched experiments.
- Implement FedMADE and SecureDyn-inspired trust, followed by poisoning and the proposed combined method.